Web Programming

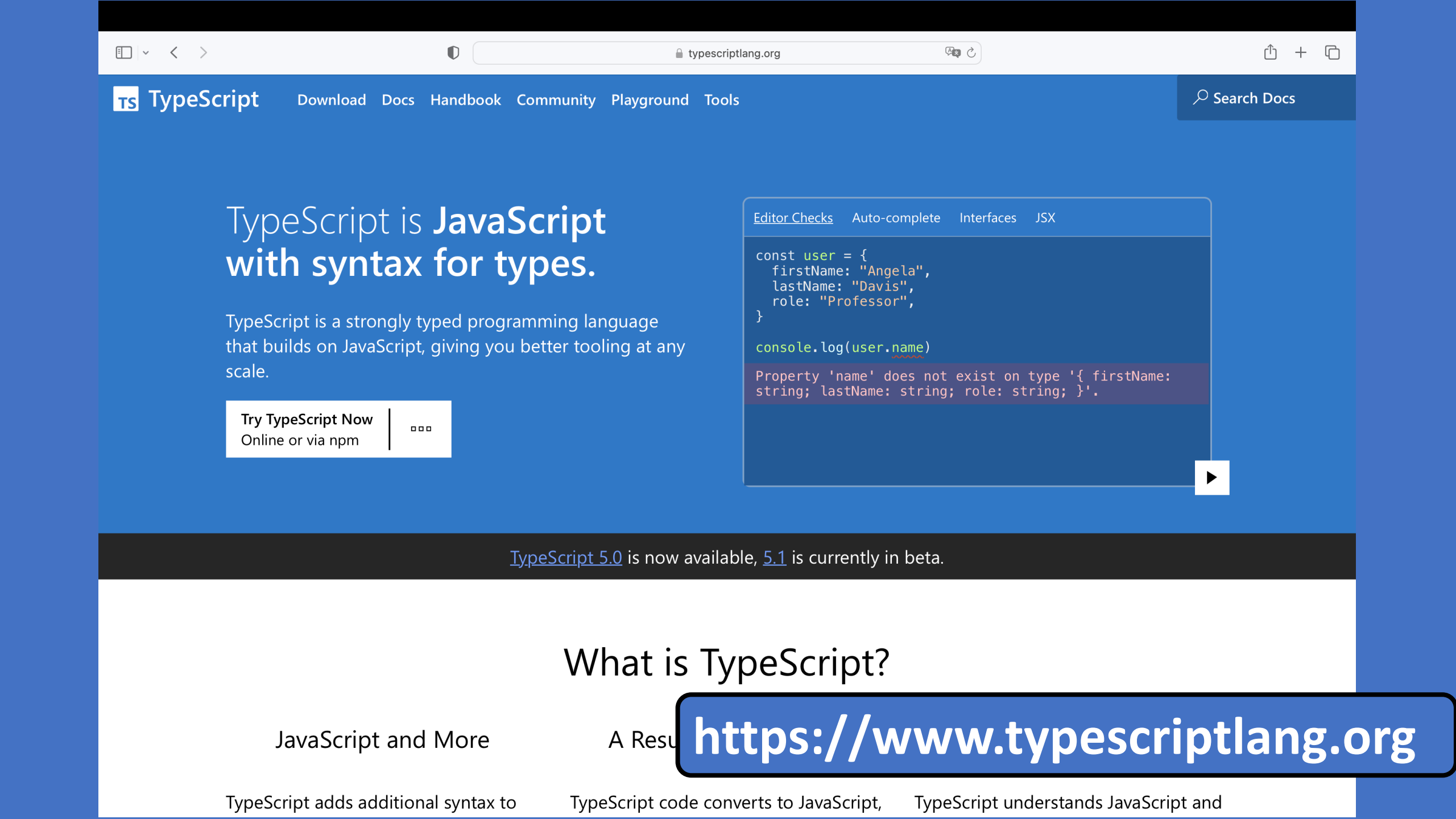
Rodrigo Martins Pagliares

Last update: 05/03/2022

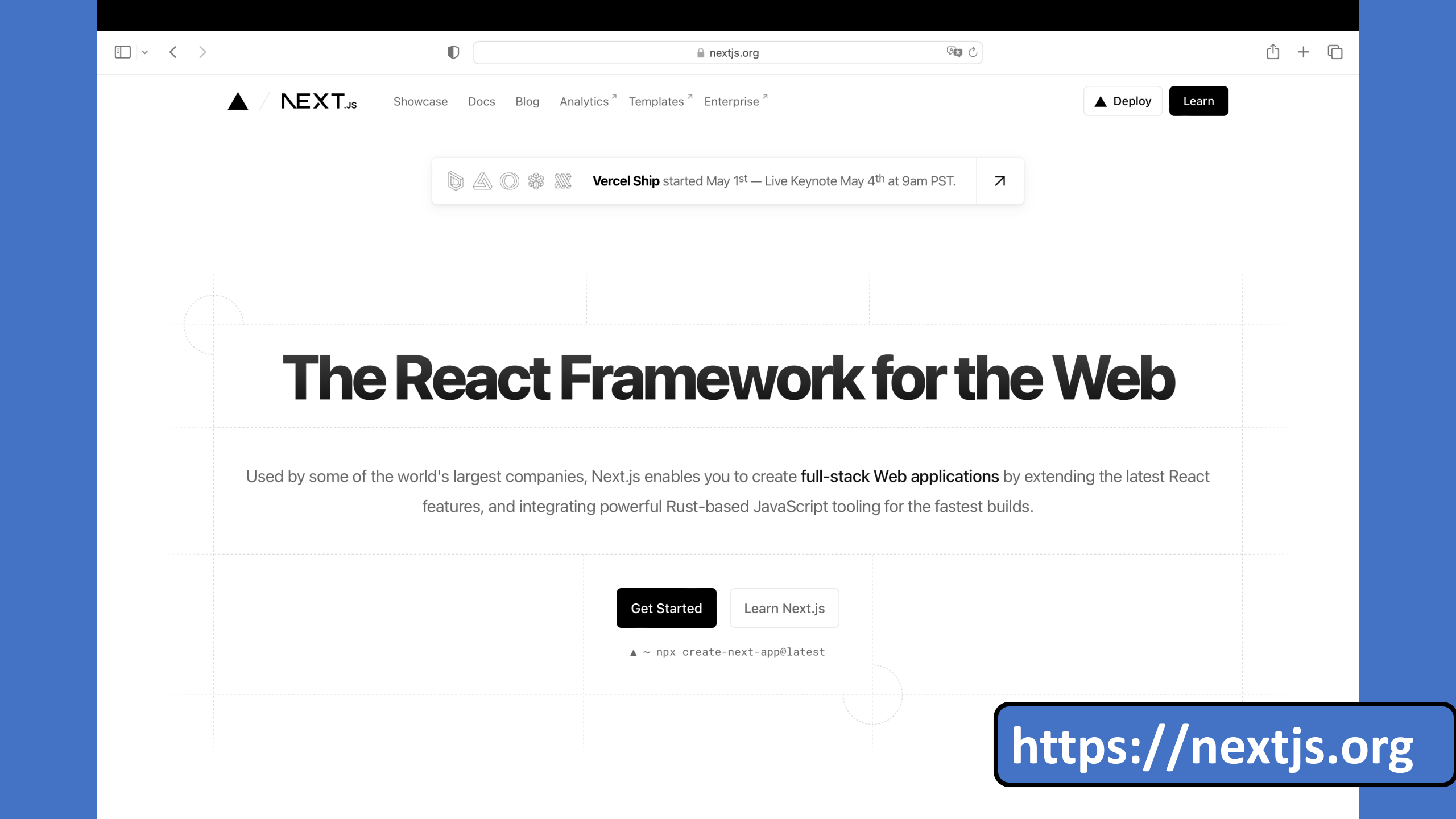
Web programming foundations

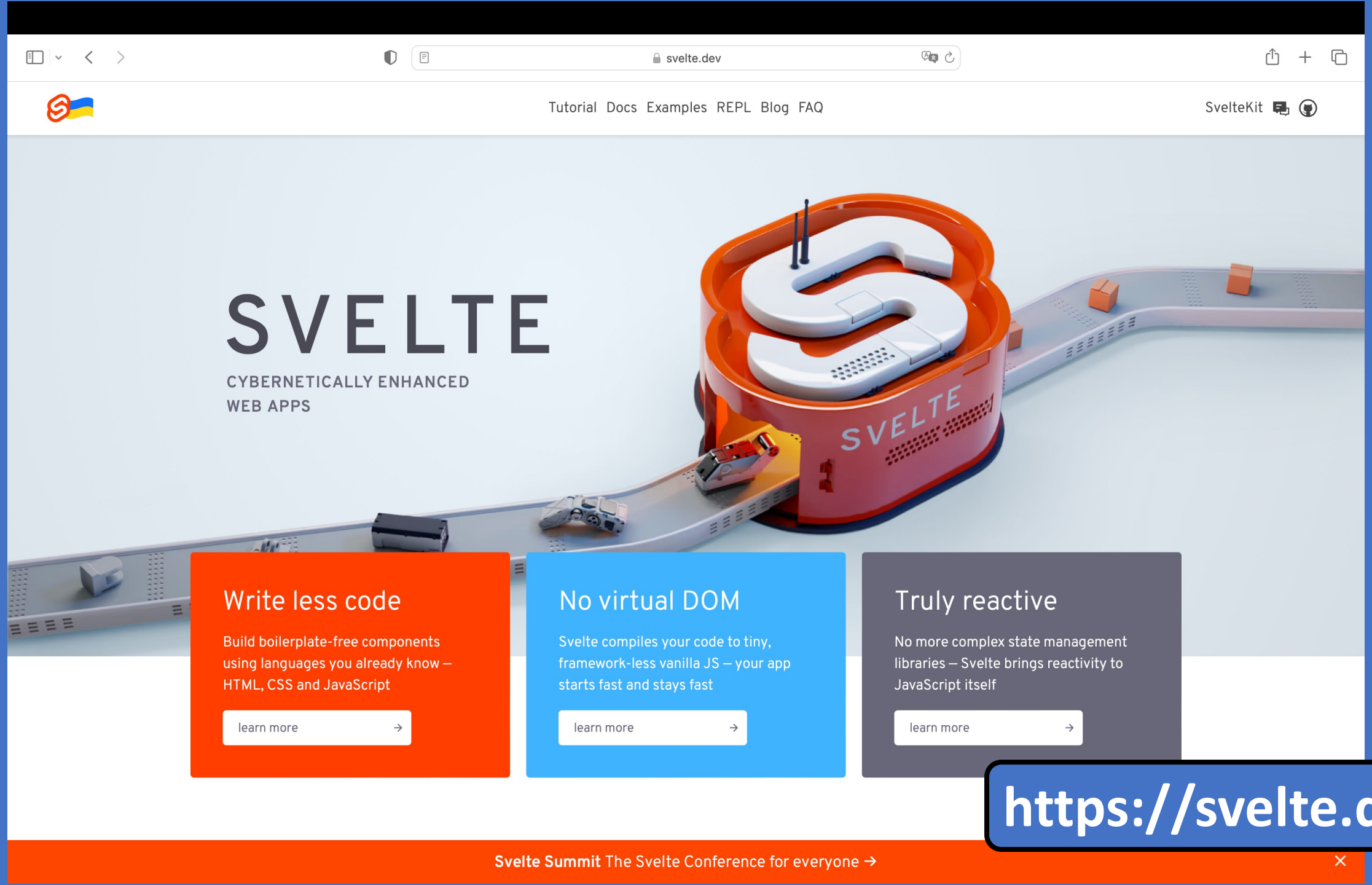
Emergent technologies

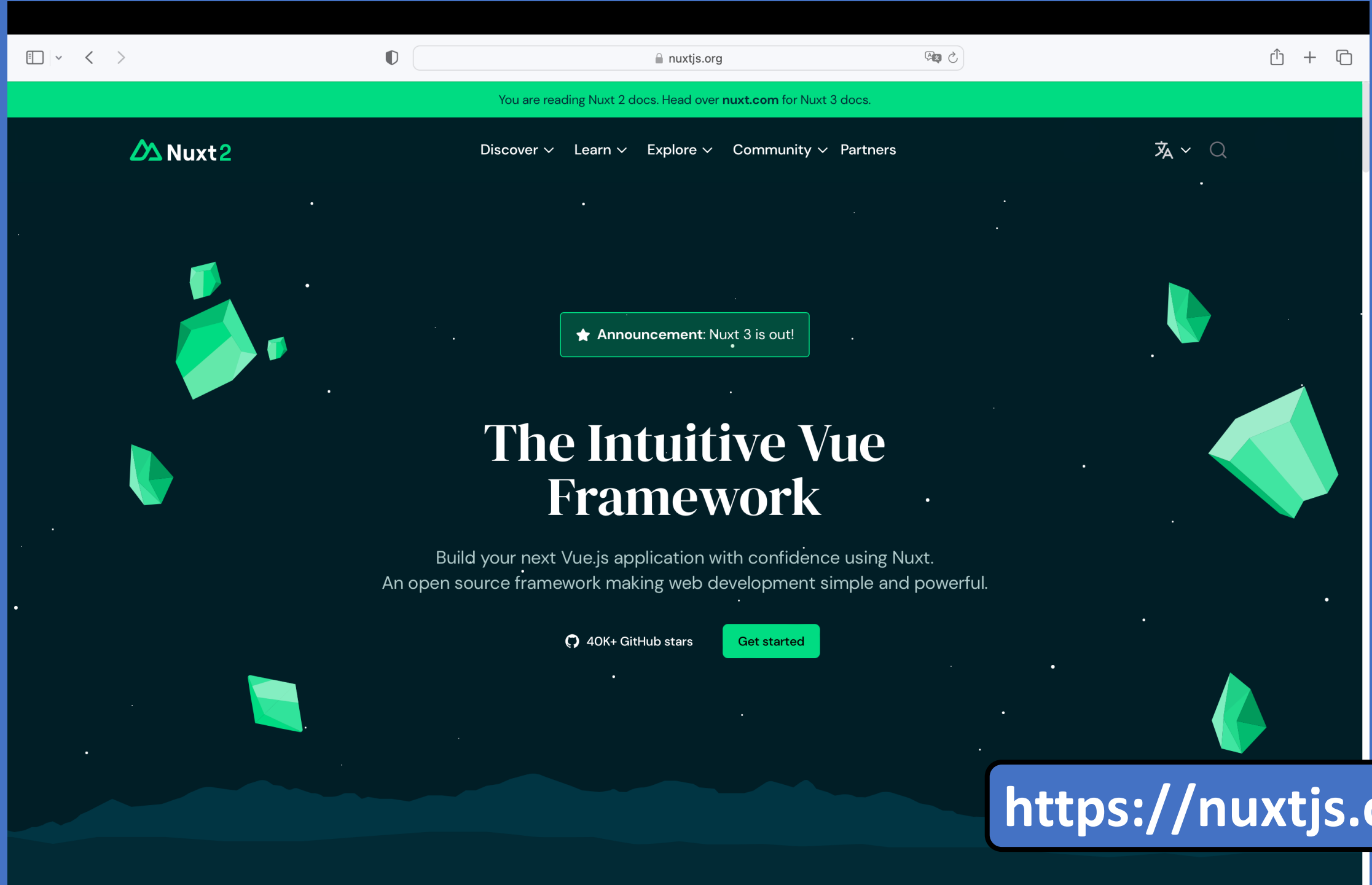
Programming, scripting, and markup languages

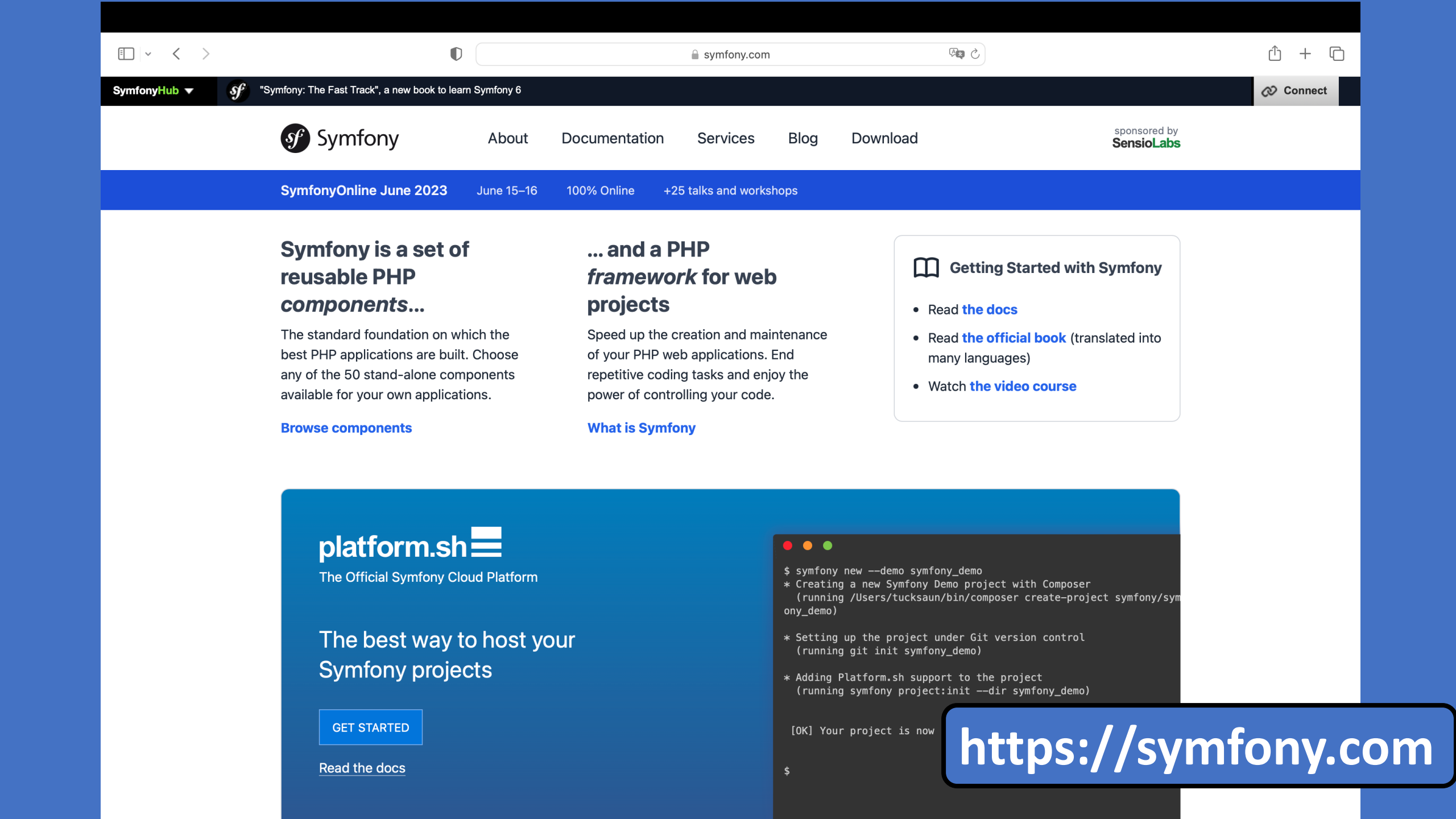

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TypeScript is JavaScript with syntax for types.
TypeScript is a strongly typed programming language that builds on JavaScript, giving you better tooling at any scale.
Try TypeScript Now
Online or via npm
Editor Checks
Auto-complete
Interfaces
JSX
const user = {
firstName: "Angela",
lastName: "Davis",
role: "Professor",
}
console.log(user.name)
Property 'name' does not exist on type '{ firstName: string; lastName: string; role: string; }'.
TypeScript 5.0 is now available, 5.1 is currently in beta.
What is TypeScript?
JavaScript and More
TypeScript adds additional syntax to
A Resu
TypeScript code converts to JavaScript,
TypeScript understands JavaScript and
https://www.typescriptlang.org

Web frameworks and technologies


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The React Framework for the Web
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Browse components
... and a PHP framework for web projects
Speed up the creation and maintenance of your PHP web applications. End repetitive coding tasks and enjoy the power of controlling your code.
What is Symfony
Getting Started with Symfony
Read the docs
Read the official book (translated into many languages)
Watch the video course
platform.sh
The Official Symfony Cloud Platform
The best way to host your Symfony projects
GET STARTED
Read the docs
$ symfony new --demo symfony_demo
* Creating a new Symfony Demo project with Composer
(running /Users/tucksaun/bin/composer create-project symfony/symony_demo)
* Setting up the project under Git version control
(running git init symfony_demo)
* Adding Platform.sh support to the project
(running symfony project:init --dir symfony_demo)
[OK] Your project is now
$
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Build rich, interactive web applications quickly, with less code and fewer moving parts. Join our growing community of developers using Phoenix to craft APIs, HTML5 apps and more, for fun or at scale.

GET STARTED

```
defmodule TimelineLive do
  use Phoenix.LiveView

  def render(assigns) do
    render("timeline.html", assigns)
  end

  def mount(_, socket) do
    Twitter.subscribe("elixirphoenix")
    {:ok, assign(socket, :tweets, [])}
  end

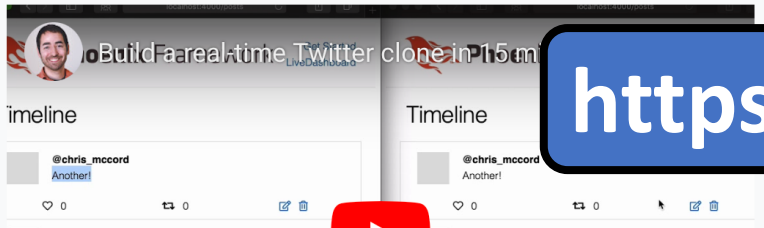
  def handle_info({:new, tweet}, socket) do
    {:noreply,
     update(socket, :tweets, fn tweets ->
       Enum.take([tweet | tweets], 10)
     end)}
  end
end
```

https://my-phx-app.com

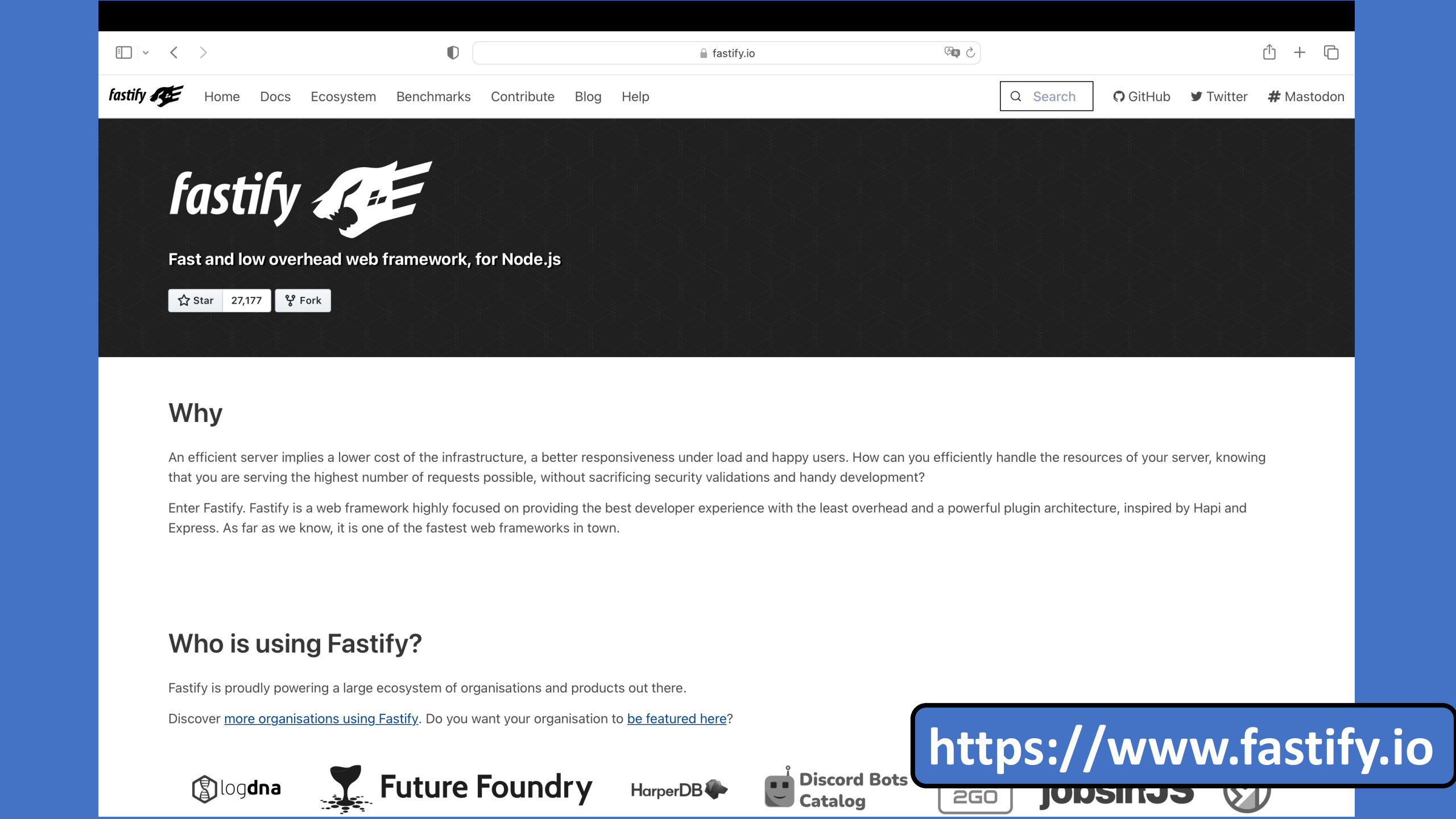
@seradio
Episode 394: Chris McCord on Phoenix LiveView

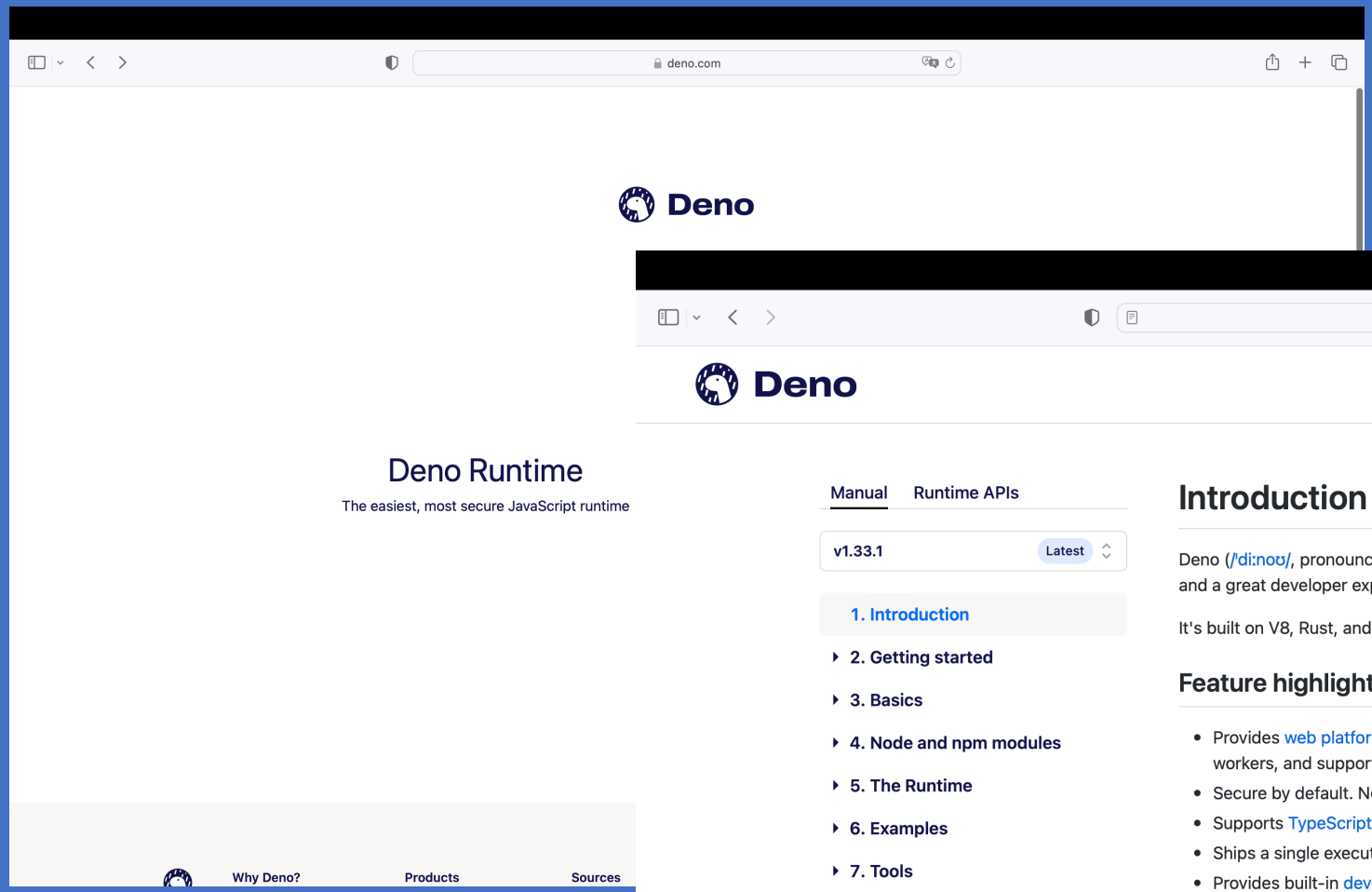
Phoenix LiveView

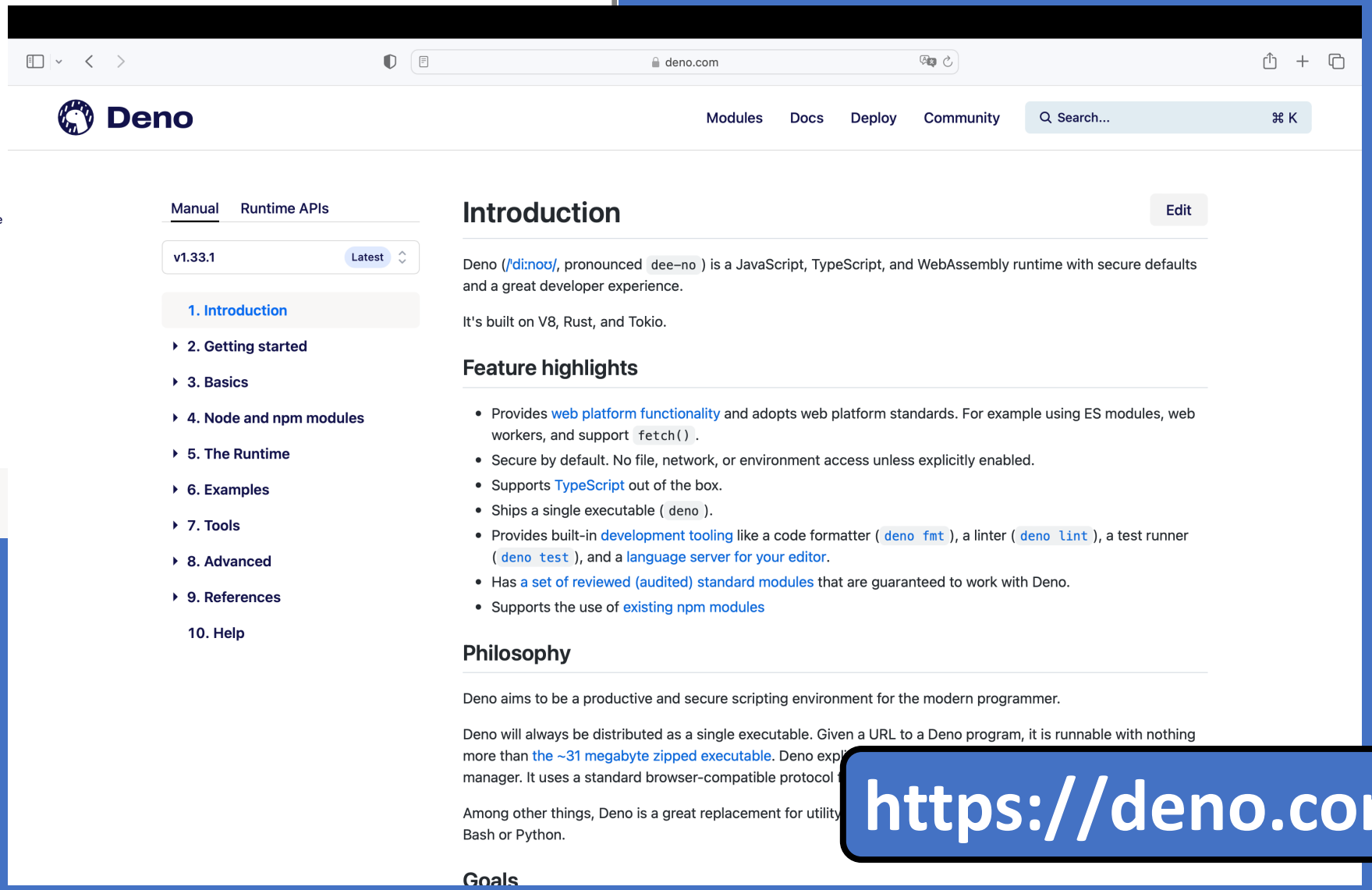
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fastify
Fast and low overhead web framework, for Node.js
Star
27,177
Fork
Why
An efficient server implies a lower cost of the infrastructure, a better responsiveness under load and happy users. How can you efficiently handle the resources of your server, knowing that you are serving the highest number of requests possible, without sacrificing security validations and handy development?
Enter Fastify. Fastify is a web framework highly focused on providing the best developer experience with the least overhead and a powerful plugin architecture, inspired by Hapi and Express. As far as we know, it is one of the fastest web frameworks in town.
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Future Foundry
HarperDB
Discord Bots Catalog
2GO
https://www.fastify.io





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play

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The High Velocity Web Framework For Java and Scala

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or browse all versions

READ THE DOCS

Documentation 2.8.19

or go to Java docs or Scala docs

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Play Framework makes it easy to build web applications with Java & Scala.

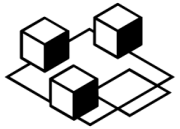
Play is based on a lightweight, stateless, web-friendly architecture.

Built on Akka, Play provides predictable and minimal resource consumption

Other frameworks and libraries

spring.io

What Spring can do



Microservices

Quickly deliver production-grade features with independently evolvable microservices.



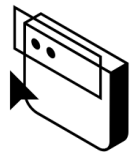
Reactive

Spring's asynchronous, nonblocking architecture means you can get more from your computing resources.



Cloud

Your code, any cloud—we've got you covered. Connect and scale your services, whatever your platform.



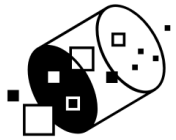
Web apps

Frameworks for fast, secure, and responsive web applications connected to any data store.



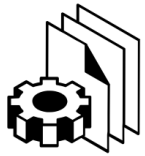
Serverless

The ultimate flexibility. Scale up on demand and scale to zero when there's no demand.



Event Driven

Integrate with your enterprise. React to business events. Act on your streaming data in realtime.



Batch

Automated tasks. Offline processi
data at a time t
you.

https://spring.io

reactnative.dev

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React Native

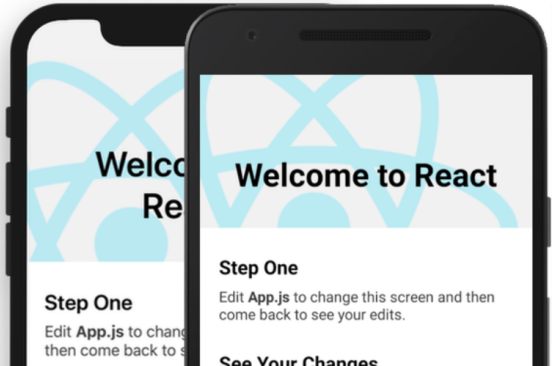
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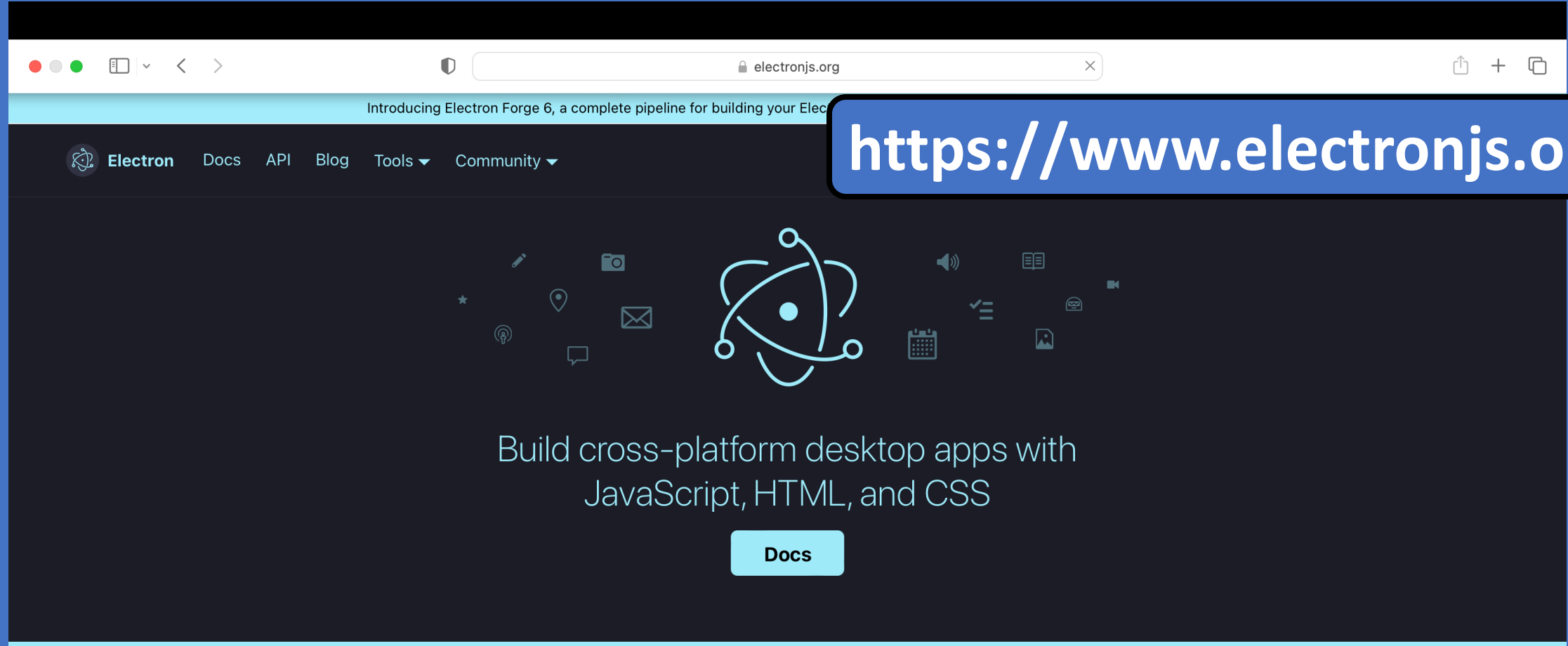
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https://reactnative.dev

https://www.electronjs.org



Web Technologies

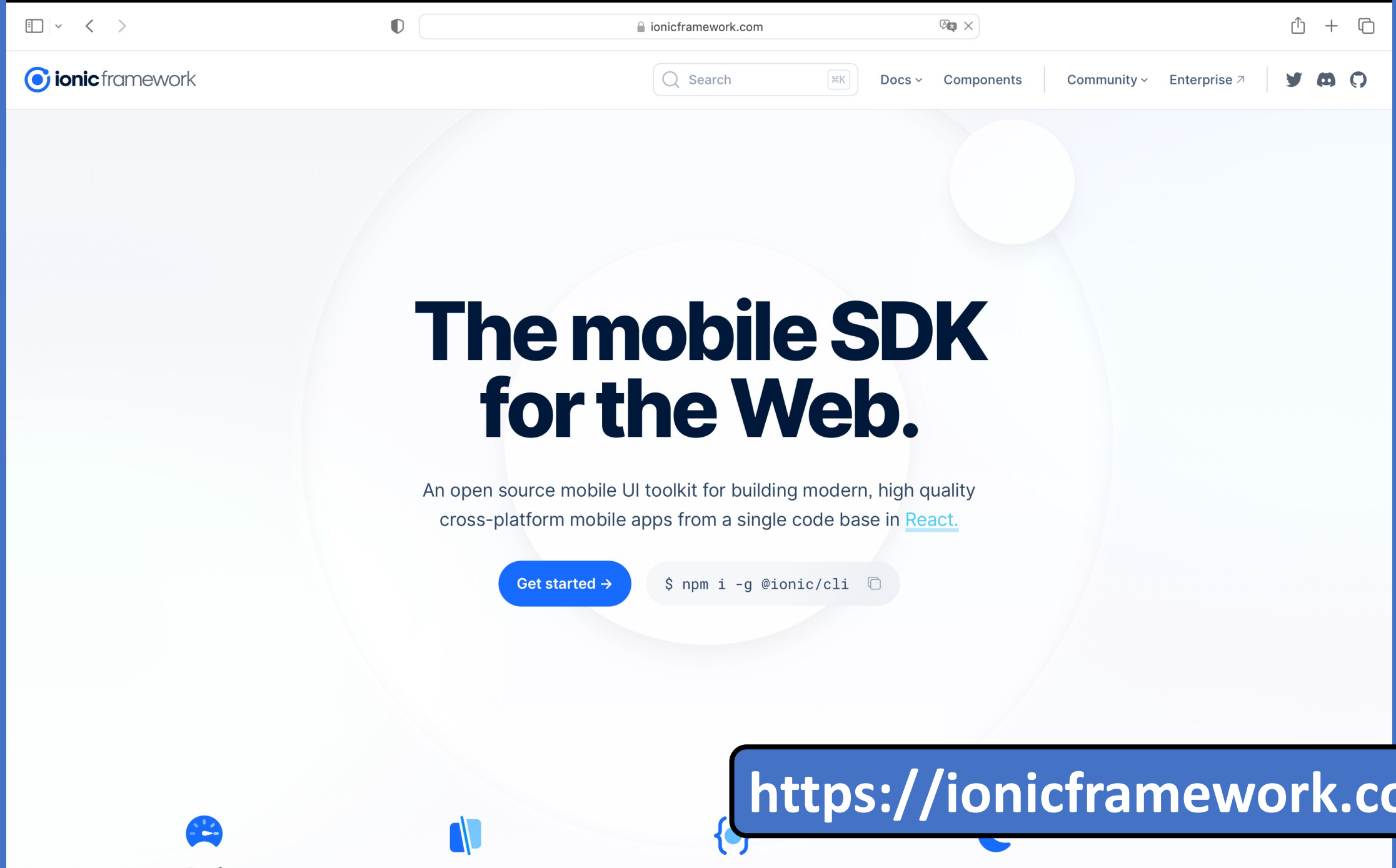
Electron embeds Chromium and Node.js to enable web developers to create desktop applications.

Cross Platform

Compatible with macOS, Windows, and Linux, Electron apps run on three platforms across all supported architectures.

Open Source

Electron is an open source project maintained by the OpenJS Foundation and an active community of contributors.


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The mobile SDK for the Web.
An open source mobile UI toolkit for building modern, high quality cross-platform mobile apps from a single code base in React.
Get started →
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https://ionicframework.com

cordova.apache.org

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Support for offline scenarios

Access native device APIs

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Other tools

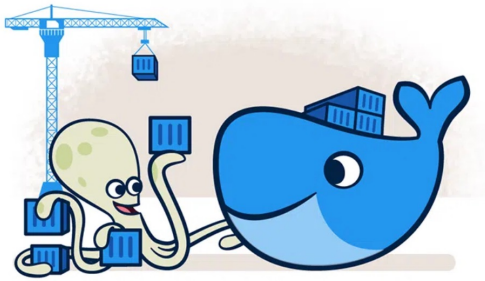
docker.com

Docker makes development efficient and predictable

Docker takes away repetitive, mundane configuration tasks and is used throughout the development lifecycle for fast, easy and portable application development – desktop and cloud. Docker's comprehensive end to end platform includes UIs, CLIs, APIs and security that are engineered to work together across the entire application delivery lifecycle.

Build

- Get a head start on your coding by leveraging Docker images to efficiently develop your own unique applications on Windows and Mac. Create your multi-container application using Docker Compose.
- Integrate with your favorite tools throughout your development pipeline – Docker works with all development tools you use including VS Code, CircleCI and GitHub.
- Package applications as portable container images to run in any environment consistently from on-premises Kubernetes to AWS ECS, Azure ACI, Google GKE and more.



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https://www.docker.com

yarnpkg.com

Important: This documentation covers modern versions of Yarn. For 1.x docs, see classic.yarnpkg.com.

yarn HOME GETTING STARTED FEATURES CLI CONFIGURATION ADVANCED API Search the documentation

Search packages (i.e. babel, webpack, react...)

The Yarn team stands with the people of Ukraine during this heinous assault on their freedom, their independence, and their lives.

To support Ukraine in their time of need visit this page.

Safe, stable, reproducible projects

Yarn is a package manager that doubles down as project manager. Whether you work on one-shot projects or large monorepos, as a hobbyist or an enterprise user, we've got you covered.



Workspaces

Split your project into sub-components kept within a single repository.



Stability

Yarn guarantees
continue to work the same way in the future.

https://yarnpkg.com

tailwindcss.com

tailwindcss

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Rapidly build modern websites without ever leaving your HTML.

A utility-first CSS framework packed with classes like `flex`, `pt-4`, `text-center` and `rotate-90` that can be composed to build any design, directly in your markup.

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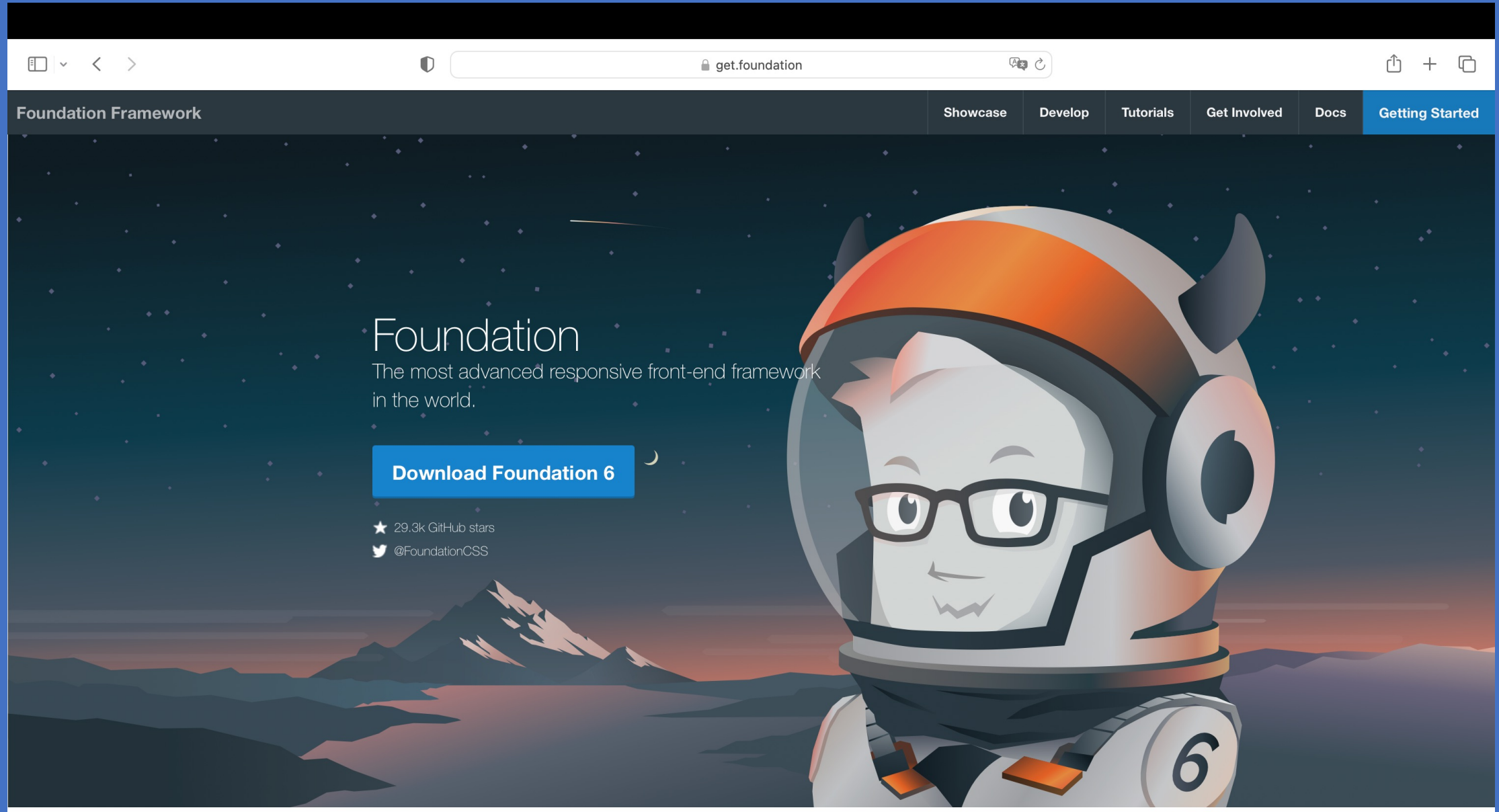
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“Tailwind CSS is the only framework that I've seen scale on large teams. It’s easy to

```
<figure class="bg-slate-100 rounded-xl dark:bg-slate-800">
  <img class="w-24 h-24" src="/sarah-dayan.jpg" alt="" width=
  <div class="pt-6 space-y-4">
    <blockquote>
      <p cl
        “T
        on
        an
      </p>
    </blockquote>
```

https://tailwindcss.com



Responsive design gets a whole

https://get.foundation

A Framework for any device, medium, and accessibility. Foundation is a family of responsive front-end

materializecss.com

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